

storage offered by the cloud to store a lot of their data assets. The time taken to run data processing and analytics jobs has dramatically reduced due to data proximity.

- **Elasticity:** Enterprises often discover that the scope and the domain of their problem shift very quickly. By the time they try to address the original issue, this may either vanish or get magnified. Optimally planning the on-premise infrastructure for such dynamic business problems could be very challenging. The elasticity offered by the cloud to scale from a few clusters to a few hundred clusters can help enterprises to be more agile and productive.
- **Better management:** Hadoop on the cloud helps enterprises to better leverage the constant innovations happening in the Hadoop ecosystem. The cloud provider is responsible to do the heavy lifting of upgrading Hadoop components while enterprises can focus on innovation.

Big Data and artificial intelligence – A match made in heaven

The world is witnessing digital disruption and transformation at an unprecedented pace, and data is at the centre of these changes. It is difficult to think of any industry that will not be transformed by AI in the years to come.

In very simple terms, AI can be thought of as a set of technologies that empower connected machines to execute tasks in a smarter way by learning, evolving and improving upon their learnings in an iterative and recursive manner. AI has been around for decades but its growth was confined due to various reasons. Today we have access to data on demand and in real-time. We have tools and technologies to perform rapid analysis. Big Data has helped to democratise AI to a larger extent. The reason Big Data has made AI popular is simply because the former can deal with massive and complex data sets in ways that traditional data processing methods (and humans) cannot.



Figure 3: Implementation of Big Data in different sectors

Big Data and AI have truly formed a great synergy, and they complement each other, bringing out the best of what either can offer individually. We need to apply AI to unlock value from Big Data and we need to feed more data to AI for it to get better and better. In short, AI is a way to navigate and unlock insights in the world of Big Data, while the latter is the way to make the former more accurate.

Before we conclude, let's look at how Big Data applications are being applied across various industry verticals.

There is hardly any sector in the digital economy that is not touched by Big Data. Across all verticals, Big Data is fuelling innovation and productivity, proving to be vital for gaining competitive advantage (Figure 3).

Since 2000, over half the Fortune

500 companies have dropped off the list or shut down entirely due to digital disruption. It has been aptly said that companies that do not understand the importance of data management will find it extremely difficult to survive in the modern digital era. Data is going to be the biggest differentiator for businesses in the years to come. Enterprises that can unlock value in this data avalanche and use it to their strategic advantage will win. Big Data is recognised as one of the most important areas of the modern data tsunami and rapid technical innovation in the Hadoop ecosystem is going to make managing this data easier. Enterprises need to ensure that they are ready to leverage Hadoop, advanced analytics and AI on Big Data to stay relevant and profitable, and to gain competitive advantage. **END** 🐧

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